Chang Li

University of Virginia xue6bq@virginia.edu +1 (434) 282-5127 linkedin.com/in/chang-li-25b112276

Education

University of Virginia (UVA) **August 2022 – Present**
Ph.D. in Statistics *Charlottesville, VA*
Research Interests:
Random graphs, Network analysis, Optimal transport, Wasserstein's distance, Statistical optimization

Shanghai Jiao Tong University (SJTU) **Septempber 2018 – June 2022**
Bachelor of Science in Mathematics and Applied Mathematics *Shanghai, CN*

Working Experience

Data Analyst Intern **February 2021 – May 2021**
Cardinal Operations *Shanghai, CN*

- Integrated sales and inventory data from multiple years and performed exploratory data analysis for Mondelēz International
- Used Python to clean, import and stucture the data, facilitating efficient analysis for simulations
- Fit parametric models for the sales and inventory data and developed functions to simulate sales and replenishment events
- Predicted the future scenario and provided actionable insights, optimizing the efficiency for the company

Research Projects

A new stable higher-order influence function estimator **September 2021 – June 2022**
Advisor: Lin Liu | School of Mathematical Sciences *Shanghai Jiao Tong University*

- Explored the generalization of first-order influence functions to higher-order influence functions (HOIFs) for rate-optimal estimators in semiparametric statistics
- Identified the practical challenges with existing empirical HOIF (eHOIF) estimators, such as numerical instability and issues with large-dimensional sample Gram matrix inversion
- Developed a more stable version of the HOIF estimator (sHOIF) that provides superior statistical guarantees

Improved Curvature Gaps for Stochastic Blockmodel Graphs **September 2023 – Present**
Advisor: Zachary Lubberts | Department of Statistics *University of Virginia*

- **Goal**: Recover the membership of edges within communities or between communities from a geometric perspective, extending the notion of curvature from differential geometry to discrete graphs (Ollivier's Ricci curvature)
- Study the curvature distributions in large SBMs to figure out when a curvature gap exists – when edges within communities have curvatures that are more positive than edges between communities, and these are well-separated
- Divide the parameter space into three signal strengths regimes and derive the distributional results using different approximations of ORC within each of them

Associations between brain connectivity and behavior outcomes **Jun 2025 – Present**
Advisor: Aiying Zhang | Department of Data Science *University of Virginia*

- **Goal**: Predict neuroimaging traits and identified associated ROIs (Regions of Interest) using brain connectivity and fMRI (functional MRI) data represented as matrices.
- Performed separate matrix regressions and used the combined solution to initialize a Lasso model, optimized via proximal gradient descent with Nesterov acceleration.
- Enhanced prediction accuracy, interpretability, and coefficient sparsity through this two-stage modeling approach.

Teaching Experience

Teaching Assistant

STAT 2120 Introduction to Statistical Analysis, University of Virginia *Fall 2023 and Spring 2024*

Instructor

STAT 1100 Chance: An Introduction to Statistics, University of Virginia *Fall 2024 and Spring 2025*

STAT 1602 Introduction to Data Science with Python, University of Virginia *Fall 2025 and Spring 2026*

Presentations and talks

"Improved Curvature Gaps for Stochastic Block Model Graphs", SIAM Conference on Mathematics of Data Science poster session, Atlanta, GA. October 21-25, 2024

"Community Detection via Curvature Gaps", CFE-CMStatistics, London, UK. December 13-15, 2025

"ConnectMVR: A Supervised Brain Connectivity Analysis Framework for Predicting Behavioral Outcomes", UVa Quantitative Psychology DADA (Design and Data Analytics), Charlottesville, VA, November 6, 2025

Skills

Programming Languages: Python, R, SQL, Matlab, SAS, Mathematica, LaTeX